Blood Metals and Metabolic Syndrome: A Machine Learning and Mendelian Randomization Investigation

1. State of Problem

Ask how someone got their high blood lead levels and the answer is usually boring: home or work. Paint chips in old houses. Soil near a road that was never tested. Pipes no one got around to replacing. None of it is new either: leaded gasoline hasn't been sold for decades, but its residue still appears in exposure data. Industrial emissions add to the mélange. This risk is further heightened in workplaces built around battery recycling or metal smelting, whereby workers bring it home, often unknowingly, and expose their families in the process [1,2].

A threshold level below which lead exposure is considered safe has not been identified [3,4]. It builds up in the body and impacts several organ systems simultaneously: neurological, cardiovascular, renal, hematological. Even low levels of exposure have been linked to impaired cognitive development, behavioral problems, increased blood pressure and cardiovascular disease, while higher levels of exposure can cause kidney damage, anemia or death [3-5]. The children get the worst of it. Their nervous system is still developing. This alone puts them at greater risk and the resulting cognitive deficits can persist well into adulthood [4,6].

Lead is the best-documented of these exposures, but it is not the only one to be concerned about. Exposure to cadmium and mercury comes mainly from cigaret smoke, shellfish consumption, and industrial exposure. Both of these metals have been shown to track fairly consistently with increased odds of metabolic syndrome and several of its individual components in large observational cohorts [7-11]. Less clear is the case for selenium, which is required by the body but where blood concentrations above the physiological range have also been associated with increased prevalence of metabolic syndrome, particularly in the presence of zinc [12-14]. Manganese is the least settled of the four, with some studies showing risk increase only in specific age groups and others none [15,16]. None of these associations behaves the same way in different populations, which is itself part of the motivation for testing all five metals side-by-side in the same cohort, rather than treating lead as a stand-in for the group.

Metabolic syndrome (MetS) is not one disease. It's a pattern: abdominal obesity, insulin resistance, high blood pressure, high triglycerides, low HDL cholesterol: all showing up together in the same patients more often than chance would predict. Each on its own increases the risk of cardiovascular disease and type 2 diabetes. But stacked together, they push that risk far beyond what you'd get by just adding the individual effects [17-21].

Most of the damage is done by insulin resistance and central obesity . Then you add on dyslipidemia and hypertension. The full mechanistic picture is not yet complete, but bad diet and lack of physical activity are clearly near the center of it [22-24].

Mortality associated with MetS, particularly cardiovascular mortality, continues to rise with its global prevalence, partly driven by parallel epidemics of obesity and physical inactivity that do not appear to be abating [20,25,26].

Blood lead is associated with MetS throughout more than one biological pathway. As lead levels increase MetS risk increases and its individual components (blood pressure, triglycerides, fasting glucose and HDL cholesterol) all go in the predicted way. A notable data point comes from the Korea National Health and Nutrition Examination Survey where doubling blood lead alone was related with a more than 1.5-fold increase in probabilities of MetS [27]. Lead also interferes with lipid metabolism and causes insulin resistance, the two pathways that are thot to be most directly responsible for MetS, and increased lead has been independently associated with increased triglycerides and fasting glucose and decreased HDL [28,29]. There is also a cardiovascular aspect: oxidative stress generated by lead seems to aggravate circulatory dysfunction and older patients with MetS seem more susceptible to autonomic issues with continuous exposure [5,30]. Thyroid function may also play a role, as thyroid-stimulating hormone has been identified as a potential mediator between lead exposure and MetS [31].

The molecular picture for the other four metals is similar: oxidative stress, systemic inflammation and altered lipid and glucose metabolism continue to emerge as the routes linking metal exposure to MetS, independent of which metal is being studied [13,32]. The difference lies in the extent to which co-exposure, genetic background, and lifestyle factors such as food and smoking bend that link in every particular group, which is likely why the amount and even the direction of the correlation vary so dramatically from one cohort to the next [33-35].

But none of this resolves the question of how strong the blood lead–metabolic syndrome link is. Several Korean studies have shown a clear association, where higher blood lead is associated with higher odds of MetS, higher triglycerides, and larger waist circumference [10,27,28]. Other studies do not see the same pattern. Findings vary across samples and age bands, and the strength of the lead association is not consistent from one study to the next, which is why the pooled estimates carry substantial heterogeneity [7,8,36]. Some data indicate that lead affects specific components of MetS (blood pressure and lipids in particular) more than it predicts the syndrome as a coherent whole [29,37].

The picture is no better for the other metals. Large cross-sectional studies and meta-analyses report inverse or null relationships for lead, cadmium and mercury in particular populations, so the direction of association is not settled even for the two metals with the most consistent overall evidence [8,16,38]. Analyses of United States populations have reported associations that differ from those in Asian cohorts in both direction and magnitude, which is a harder problem than a simple difference in effect size [33,34]. Manganese is the least clear of the group, with higher risk seen in some studies and only in certain age groups, and not seen at all in others [15,16].

Machine learning paired with SHAP (SHapley Additive exPlanations) has turned into a go-to combination for disease prediction. It is also useful for something more specific: figuring out which risk factors actually matter, and by how much. XGBoost, Random Forest, and Gradient Boosting are the usual suspects here. They have already been run against hypertension, heart disease, Alzheimer's disease, and

diabetic retinopathy: different diseases, same basic move, pulling together clinical, demographic, and lifestyle data to reach high predictive accuracy [39-42].

What ML does particularly well is rank risk factors rather than just flag them. SHAP scores each feature's contribution within a model, so researchers can see exactly which variables are doing the predictive work: cholesterol, blood pressure, and age for heart disease; hemoglobin A1c and systolic blood pressure for diabetic retinopathy [40,43,44]. The same technique visualizes how, say, age and triglycerides feed into hypertension risk, which in turn opens the door to more targeted interventions [39,45].

That interpretability matters clinically, because it is often the missing ingredient that keeps complex models out of practice. ML-SHAP combinations have already been used to predict pressure injuries in ICU patients and to assess COVID-19 reinfection risk, offering actionable, early-warning insight in both cases [46,47].

Mendelian Randomization (MR) offers a way around the uncertainty that plagues observational studies. By using genetic variants as instrumental variables, it can test causal links between a risk factor and a health outcome without needing a randomized controlled trial, which, for something like metal exposure, simply is not feasible or ethical. Because genetic variants are assigned essentially at random during meiosis, they are distributed independently of the confounders that trouble observational designs. Using a genetic proxy for an exposure (cholesterol levels being a classic example) lets researchers check whether an observed association actually reflects a causal effect, or just correlation [48-50].

Taken together, the literature points to a plausible but unsettled role for blood metals (lead foremost among them, but cadmium, mercury, selenium, and manganese close behind) in the development of metabolic syndrome. Untangling which of these associations reflect real biology, and which reflect confounding by diet, smoking, or socioeconomic circumstance, will take more than another single-metal cross-sectional study. This proposal takes a two-step approach to that problem. First, machine learning ranks all five metals, alongside a broad panel of demographic, anthropometric, and lifestyle covariates, by how much each actually contributes to MetS risk once the others are accounted for, rather than assuming in advance which one matters most. Second, Mendelian Randomization takes whichever exposures come out of that ranking with a defensible genetic instrument (the metals, and any non-metal factor the modeling flags as comparably important, such as body mass index) and tests whether their contribution reflects a genuine causal effect or just confounding.

2. Literature Review

Age, sex, and gender all shape MetS risk. Nobody disputes that much. What is harder to pin down is exactly how they interact, since the pattern shifts by population and by study design. Prevalence rises with age in both sexes, and older adults consistently show higher rates than younger ones [26,51-53], though the shape of that rise differs by sex. Before age 50, men in many populations run a higher MetS prevalence than women; after menopause, women's rates climb sharply and often overtake men's

[54,55]. Hormonal shifts explain part of this reversal. So do gender-related factors like lifestyle and psychosocial stress [56,57].

The way MetS components cluster also splits along sex lines. Abdominal obesity tends to dominate the picture for women, whereas men's risk profiles are more mixed, often skewing toward elevated blood pressure and triglycerides at younger ages [52,54,56]. Age hits women harder in odds-ratio terms: several studies report a steeper age-related climb in MetS odds for women than men [54,55], and older women appear to carry a heavier burden of the cardiometabolic and cardiovascular consequences that follow [51,58].

Not everyone agrees on the shape of these trends, though. Some studies find similarly steep age-related increases for both sexes; others report that the sex gap narrows, or even flips, in the oldest cohorts [51,58]. Which components drive the risk can shift too: triglyceride disorders tend to ease with age in men while holding steady or worsening in women [54,58]. And there is a deeper methodological question underneath all of this: whether applying the same diagnostic cut-offs across every age and sex group even makes sense, given how differently individual components seem to behave across demographic subgroups [17,59].

Age, sex, and gender clearly matter for MetS risk, but the biological, behavioral, and social forces behind that relationship pull in different directions depending on the population studied. That argues for age- and sex-specific approaches to diagnosis and prevention rather than one-size-fits-all criteria, and for more work aimed at resolving the inconsistencies that keep showing up across studies [51,52,54,56,58].

Race/ethnicity, education, marital status, household size, and the poverty–income ratio each leave a mark on MetS risk, though not always in a straightforward or consistent way. Non-Hispanic Black, Hispanic, and some Asian subgroups tend to show higher MetS prevalence and severity than non-Hispanic Whites even once socioeconomic and lifestyle factors are accounted for [52], but the pattern is not uniform within groups either. Non-Hispanic Black men, for instance, sometimes show lower MetS prevalence than White men, while Black women show higher rates than White women [52]. These disparities persist across the life course, and environmental or behavioral factors alone do not fully explain them [52,60].

Socioeconomic status generally tracks with MetS risk the way you would expect: lower education and income mean higher risk, particularly for women and certain ethnic groups [60]. But the relationship bends depending on who you are looking at. Higher income protects non-Hispanic Whites and Asians from MetS, yet among non-Hispanic Blacks it is sometimes neutral, or even linked to higher obesity rates [52,60]. Curiously, how people perceive their own social standing predicts MetS severity better than their actual income or education does, especially for women [57,61].

Household-level social variables are harder to pin down than income or education. The evidence base for factors such as marital status and household composition is thin and inconsistent, and this proposal therefore carries them as descriptive covariates rather than as hypothesised risk factors. What does replicate is that where people place themselves on a social ladder tracks metabolic risk over and above their measured income and education, and does so most strongly in women [57,61].

Lower education, lower income, and certain racial or ethnic backgrounds line up fairly reliably with higher MetS risk, but subgroup, sex, and context all bend the size and even the direction of that relationship. Prevention strategies built around a single population-wide template are likely to miss a lot of this variation [52,57,60].

BMI and hip circumference remain the default anthropometric tools in MetS research. Both track reliably with MetS as they rise, along with individual components like hypertension, dyslipidemia, and hyperglycemia [17,62]. BMI alone keeps underperforming central obesity measures such as waist circumference or waist-to-hip ratio, though, largely because it cannot tell where the fat actually sits [62].

Fold hip circumference into a ratio or composite index, though, and its predictive power improves: waist-to-hip ratio and the body adiposity index both outperform plain BMI at identifying MetS risk, especially in older adults and women [62,63]. On its own, hip circumference has occasionally shown an inverse relationship with metabolic risk, possibly because gluteofemoral fat is somewhat protective, but that finding does not hold everywhere [64,65].

Which anthropometric measure predicts MetS best is still an open question. Some studies find BMI and hip circumference (or their derived indices) roughly equivalent; others find central obesity measures clearly superior, particularly in men and certain ethnic groups [62-64]. Cut-off values complicate things further, since the optimal thresholds for BMI and hip circumference shift by age, sex, and ethnicity [59,62]. The point applies across populations and not only across individuals: thresholds validated in European-ancestry samples misclassify cardiometabolic risk when applied unchanged to Asian populations, where risk begins to rise at a lower body mass index [59].

BMI and hip circumference both associate with MetS, but how well either one predicts it depends heavily on context. Central obesity measures, and composite indices that combine waist and hip data, generally discriminate MetS risk better. Whether hip circumference itself reads as protective or risk-enhancing seems to come down to how it is measured and in whom, a reminder that anthropometric screening probably needs to be tailored rather than standardized [59,64,65].

Smoking, alcohol, and physical activity relate to MetS in ways that do not fit a single pattern: some links hold up consistently across populations, others do not.

Physical activity is the clearest signal of the three. Low or absent activity shows up repeatedly across large studies and meta-analyses as an independent risk factor for MetS, while people who are consistently or newly active carry lower risk [23,24,66]. That protective effect holds across sexes and age groups, and more active people tend to show better numbers across the board: lower blood pressure, triglycerides, and waist circumference [23,24].

Smoking is murkier. Control for other lifestyle factors and the smoking–MetS association disappears in some studies, particularly among women [67]. Other research, especially in male samples, finds the opposite: smoking raises MetS risk along with specific components like low HDL, high triglycerides, and abdominal obesity [67,68]. Pair smoking with low physical activity and the risk climbs further. How much depends on intensity and which subgroup you are looking at [67,68].

Alcohol is the messiest variable of all. Plenty of studies find no significant alcohol–MetS link [69]. Others report that heavy or frequent drinking raises risk, while light-to-moderate drinking looks neutral or even protective [69]. Differences in drinking pattern, in the sex composition of the samples, and in how the reference group is defined plausibly account for much of that disagreement, which makes population-level summaries of alcohol and MetS difficult to interpret [67,69].

Physical activity is the one lever here that behaves predictably and is genuinely modifiable; smoking and alcohol swing around depending on sex, intensity, and lifestyle context. Any prevention strategy probably needs to lean hard on activity promotion while treating smoking- and alcohol-related interventions as population-specific problems rather than universal ones [23,68,69].

Heavy metals complicate the picture further. Lead, cadmium, mercury, selenium, and manganese have all been studied against MetS, and again the findings split between consistent and contradictory.

Cadmium and mercury show the most consistent pattern: most meta-analyses and large observational studies link higher blood levels of both to greater MetS risk and to specific components like abdominal obesity, hypertension, and dyslipidemia [7-11,70]. Selenium is an odd case: it is an essential nutrient, yet elevated blood levels, especially alongside zinc, still track with higher MetS prevalence [12,13,35]. Lead shows up too, with some studies linking it to increased MetS risk, particularly in men, and to higher blood pressure, triglycerides, and waist circumference [10,27,29,36].

Several large cross-sectional studies and meta-analyses report inverse or null associations for lead, cadmium, and mercury in some populations, so the direction of association is not settled [8,16,38]. United States analyses have reported associations differing from those in Asian populations in direction as well as magnitude, rather than weaker versions of the same effect [33,34]. Manganese is the least settled of the group: increased risk shows up only in specific age brackets in some studies, and not at all in others [15,16].

The mechanistic story that has emerged points to oxidative stress, inflammation, and disrupted lipid and glucose metabolism as the pathways connecting these metals to MetS [13,32,71]. But co-exposure to multiple metals, genetic background, and lifestyle factors like diet and smoking all modify the relationship, which is probably why results look so population-specific [33-35]. Sex differences complicate things further: some metals associate more strongly with MetS in men than in women [14,70].

Cadmium, mercury, and selenium carry the most consistent evidence for increased MetS risk; lead and manganese remain mixed and heavily context-dependent. Population, exposure level, and confounding all seem to be doing real work behind that variability, which is exactly the gap prospective and mechanistic studies still need to close [7,8,72].

3. Method

3.1. Machine Learning

3.1.1. Data sources

Six consecutive NHANES cycles feed this study: 2007–2008, 2009–2010, 2011–2012, 2013–2014, 2015–2016, and 2017–2018. Pooled together, they use a stratified, multistage probability design built to produce nationally representative estimates for the U.S. civilian, non-institutionalized population. Merging the examination, laboratory, and demographic files across all six cycles gives a pooled sample of roughly 59,842 individuals. That number drops once the analysis is restricted to adults aged 20 and older, down to around 34,770 participants. From there the exclusions start. Anyone missing one of the five MetS components (waist circumference, fasting glucose, HDL cholesterol, triglycerides, or systolic/diastolic blood pressure) drops out first, which brings the cohort to approximately 13,337. From that remainder, anyone with no blood-metal measurement at all (none of lead, cadmium, mercury, selenium, or manganese recorded) also drops out, since the exposure panel requires at least one measured metal per participant; a participant with, say, lead and cadmium measured but not the other three is retained. This leaves a final analytic cohort of approximately 11,144 participants. Every analytic record carries valid sampling weights and design variables, so the unweighted and weighted cohorts come out identical in size: n ≈ 11,144, shown in Figure 1 and itemized step by step in Table 2.

For weighted analyses, the two-year MEC weight (WTMEC2YR) is used for most descriptive statistics, adjusted for pooling six cycles together following standard NHANES guidance; for the five blood metals specifically, the cycle-appropriate weight is applied instead of WTMEC2YR throughout, because two of the six cycles measured the metals panel in only a half-subsample of eligible participants. Verified directly against each cycle's own NHANES laboratory documentation: 2007–2008, 2009–2010, 2011–2012, and 2017–2018 use WTMEC2YR for the metals; 2013–2014 and 2015–2016 use the half-subsample weight WTSH2YR instead, per that documentation's explicit instruction to do so. Standard errors for the descriptive comparisons in section 3.1.4 account for NHANES's stratified, multistage sampling design via Taylor-series linearization (the "ultimate cluster" method: between-primary-sampling-unit variance of PSU-level weighted totals, summed within and across strata), using the primary sampling unit (SDMVPSU) and stratum (SDMVSTRA) design variables directly, rather than a simplified weighted-mean approximation that ignores clustering. This is the standard approach recommended in the NHANES Analytic and Reporting Guidelines for a masked, two-PSU-per-stratum public-use design. The cleaned datasets, both weighted and unweighted, are exported for downstream modeling.

Figure 1. Data-Cleaning Pipeline for the Machine-Learning Analysis

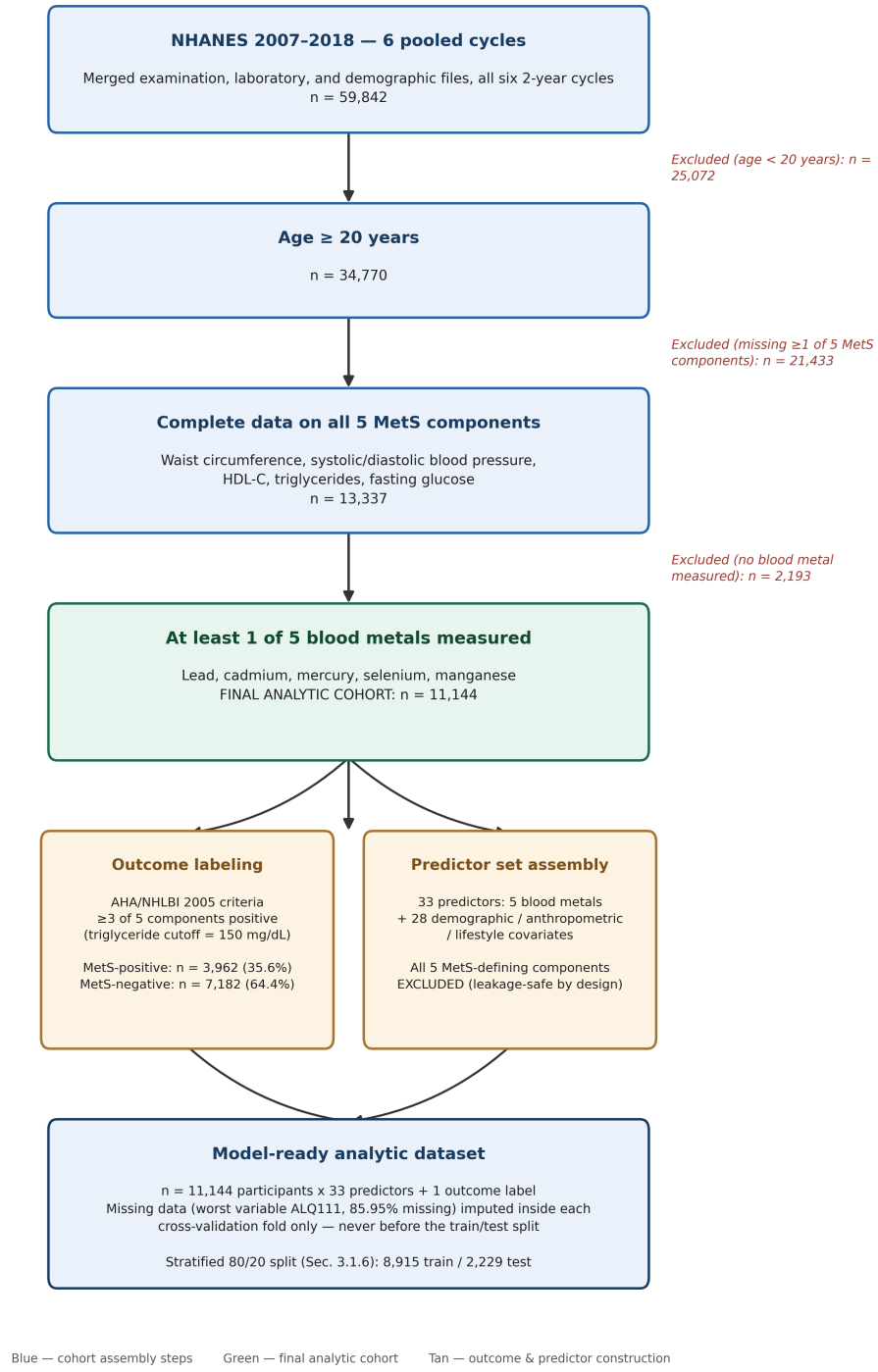


Figure 1. Data-cleaning pipeline used for the machine-learning analysis.

3.1.2. Diagnostic criteria

AHA/NHLBI (2005) thresholds, harmonized with the drug-treatment clause from the 2009 joint interim statement (Alberti et al., Circulation 2009;120:1640–1645, building on Grundy et al., Circulation 2005;112:2735–2752), set the cutoffs here. Each of the five MetS components becomes a yes/no flag on an OR basis: a component is scored positive if the lab/exam threshold is crossed: abdominal obesity (waist circumference ≥ 102 cm in men, ≥ 88 cm in women), hypertriglyceridemia (triglycerides ≥ 150 mg/dL), low HDL-C (< 40 mg/dL in men, < 50 mg/dL in women), elevated blood pressure (systolic ≥ 130 mmHg or diastolic ≥ 85 mmHg), or impaired fasting glucose (≥ 100 mg/dL), OR if the participant reports current drug treatment for that condition: antihypertensive medication (BPQ050A) for the blood-pressure component, cholesterol-lowering medication (BPQ100D) for both the HDL and triglyceride components (NHANES asks only one cholesterol-medication question across all six cycles, so it is applied to both lipid components, consistent with standard practice in the NHANES MetS literature), and insulin or oral diabetes medication (DIQ050, DIQ070) for the glucose component. Sum the flags for each participant, apply a threshold of three or more positive criteria, and MetS status gets assigned automatically, a classification step that stays transparent and fully reproducible within the pipeline. All four medication variables are used only to build this label and, like the five lab/exam components themselves, are excluded entirely from the predictor set (section 3.1.3) to avoid outcome leakage. Applying the drug-treatment clause reclassifies 1,201 of 11,144 participants from MetS-negative to MetS-positive under the lab-only definition (0 participants move the other direction, as the OR logic requires), raising the analytic cohort's MetS prevalence from 35.55% (lab/exam criteria alone) to 46.33% (harmonized definition). Both definitions are retained in the analytic dataset, and a dedicated sensitivity analysis (section 3.1.7) re-fits the top-ranked models against the lab-only label to quantify how much of the headline result depends on this definitional choice.

3.1.3. Variable definition and predictor selection

The predictor set is built to avoid a specific problem: several of the candidate risk factors discussed elsewhere in this proposal are also among the five variables used to define MetS status itself. Waist circumference, blood pressure, HDL cholesterol, triglycerides, and fasting glucose are pulled from NHANES for one purpose only: determining who is and is not classified as having MetS. None of them re-enter the model as predictors. Including them as inputs to a model that predicts a label built from those same five variables would let the model partially reconstruct its own answer key rather than learn a genuine risk relationship. Avoiding that circularity is a deliberate design choice throughout this pipeline, not an afterthought.

The predictor set instead draws on four categories, pulled directly from the NHANES tables via the preprocessing script. Blood metals anchor the exposure side: lead (LBXBPB), cadmium (LBXBCD), total mercury (LBXTHG), selenium (LBXBSE), and manganese (LBXBMN), all measured in the same PBCD laboratory module. Demographic and socioeconomic covariates cover age (RIDAGEYR), sex (RIAGENDR), race/ethnicity (RIDRETH1), education level (DMDEDUC2), marital status (DMDMARTL), household size (DMDHHSIZ), the income-to-poverty ratio (INDFMPIR), and examination month (RIDEXMON, included because blood lead and several other analytes show a modest seasonal pattern in NHANES). Body mass index (BMXBMI) is the sole anthropometric predictor; hip circumference was considered and dropped after a collinearity screen (run on the training partition only, after the train/test split, so the screen itself

cannot leak test-set information into a predictor-selection decision) found it correlated with BMI at r = 0.937 and missing in 82.0% of the training partition, making it redundant rather than informative. Lifestyle and behavioral covariates round out the set: smoking history and current intensity (SMQ020, SMQ040), alcohol consumption (ALQ_EVER, ALQ130), self-rated diet quality (DBQ700), vigorous work activity, vigorous recreational activity, and moderate recreational activity (PAQ605, PAQ650, PAQ665), active-transportation minutes (PAD645), and resting pulse (BPXPLS). ALQ_EVER is a harmonized alcohol-use gatekeeper variable rather than a single raw NHANES field: the survey's "ever consumed alcohol" question changed codes mid-series (ALQ101 in 2007–2016, ALQ111 in 2017–2018 only, confirmed via non-null counts per cycle), so the two are combined into one 0/1 indicator to avoid what would otherwise be 82.6% structural missingness in this analytic cohort from treating ALQ111 alone as a six-cycle predictor. Active-transportation minutes (PAD645) is similarly cleaned against its own NHANES skip pattern before imputation: participants who answer "No" to the gating question on walking or bicycling for transportation (PAQ635) are correctly coded as zero minutes rather than left as missing, since NHANES's questionnaire routes them past PAD645 entirely. Records without a defined MetS status are dropped; categorical variables are one-hot or ordinally encoded depending on whether the underlying categories are unordered or ranked; continuous variables are median-imputed and standardized, each paired with an always-present binary indicator flagging whether that value was actually observed (so structural or systematic missingness, for example selenium and manganese not being measured at all in the 2007–2010 cycles, remains visible to the model as information rather than being silently absorbed into an imputed constant); and all of this happens only inside the model-fitting step described in section 3.1.6, never beforehand, so that no information from a held-out fold or the test set leaks into the values used to train on the rest of the data.

3.1.4. Baseline analysis

The baseline analysis follows the NHANES complex-survey protocol without deviation. After loading the pooled 2007–2018 dataset with the cycle- and variable-appropriate weight (WTMEC2YR generally; WTSH2YR for the two blood-metals half-subsample cycles, per section 3.1.1), survey-weighted domain means are computed for each continuous variable (age, BMI, waist circumference, systolic and diastolic blood pressure, HDL-C, fasting glucose, and all five blood metals) separately for the MetS and non-MetS groups, with standard errors and between-group Wald tests obtained via Taylor-series linearization (the ultimate-cluster method) using the primary sampling unit (SDMVPSU) and stratum (SDMVSTRA) design variables, not a renormalized-weight approximation that ignores clustering, and not a simple-random-sample variance formula applied naively to a subgroup, but a proper domain (subpopulation) estimator in which non-domain-member rows still contribute their primary sampling unit's structure to the variance calculation. For the categorical covariates (sex, race/ethnicity, education, and marital status), weights are summed within each level and MetS status to produce weighted percentage shares (each with its own Taylor-linearized standard error), and an unweighted chi-square test on weight-rescaled contingency tables provides a descriptive, though not fully design-corrected, association test; a formal Rao-Scott-corrected categorical test is left to a dedicated survey package (e.g., R's `survey`) if the target journal requires it. Staying weight- and design-adjusted throughout keeps the descriptive comparisons faithful to NHANES's multistage sampling design and its varying, cycle-dependent selection probabilities.

3.1.5. Feature selection

Feature selection here is knowledge-driven rather than algorithmic: there is no LASSO or stepwise procedure deciding which variables enter the model. Every candidate predictor described in section 3.1.3 is chosen ahead of time on clinical and epidemiological grounds, then carried through imputation, standardization, and categorical encoding as a fixed set. What SHAP and permutation importance contribute afterward is a ranking, not a selection: they quantify how much each already-chosen predictor actually moves the trained model's output, and how consistently, but they never add or remove a variable from consideration. Two importance measures are computed independently and then compared against each other (SHAP values from the best-performing tree-based model by test AUC, and model-agnostic permutation importance across the top several models), so that a feature judged important because of a genuine, robust signal in the data can be distinguished from one that only looks important to a single algorithm's particular quirks.

3.1.6. Modeling

The cleaned NHANES cohort is split 80/20 into training and hold-out test sets, stratified on MetS status, to check out-of-sample performance; the test set is set aside immediately after cleaning and is not touched again until final evaluation. Eleven algorithms are trained: logistic regression, decision tree, random forest, gradient boosting, LightGBM, XGBoost, CatBoost, support vector machine, Gaussian Naive Bayes, multilayer perceptron, and k-nearest neighbors, each predicting the binary, medication-adjusted MetS label from the panel of metal, demographic, anthropometric, and lifestyle predictors described above, never from the biomarkers or medication-use variables that define MetS itself. Hyperparameters for each algorithm are tuned by grid or randomized search under stratified 10-fold cross-validation on the training partition only, scored on ROC AUC. Class imbalance (46.33% MetS prevalence under the harmonized outcome definition) is handled with SMOTENC rather than plain SMOTE, because the feature matrix mixes continuous measurements with one-hot and binary categorical variables; SMOTENC generates synthetic continuous values by interpolation as SMOTE would, but assigns categorical values by majority vote among nearest neighbors instead of interpolating them, which avoids manufacturing fractional, non-existent category values (e.g., a synthetic "sex" of 0.4) that plain SMOTE would otherwise produce on a mixed feature space. SMOTENC is refit fresh inside every cross-validation fold and inside the final training fit, never applied once to the whole dataset beforehand, so that synthetic minority-class samples are always generated from training data alone. Each continuous predictor also carries an always-present, separately-fit missingness indicator (see section 3.1.3) alongside its median-imputed value, so that structurally or systematically missing measurements remain visible to the model as information. The decision threshold used for each model's sensitivity/specificity trade-off is chosen from out-of-fold predictions on the training partition via the Youden index, not from the test set, keeping every step that could leak test-set information out of the process that produces the final, once-touched evaluation. Running eleven algorithms (from simple and fully interpretable through complex ensembles) and testing all of them on the same untouched 20% of subjects means the resulting AUC, calibration, and error metrics reflect genuine generalization rather than an artifact of any one algorithm's fit to this particular sample.

3.1.7. Model evaluation and interpretation

Evaluation runs entirely in Python (3.10+), using scikit-learn, imbalanced-learn, LightGBM, XGBoost, CatBoost, and SHAP, chosen for reproducibility and because the leakage-safe pipeline described above depends on scikit-learn's Pipeline object refitting preprocessing and SMOTENC inside every fold, a guarantee that is harder to enforce across a mixed-language workflow. Feeding the 20% test set into each fitted pipeline generates ROC curves and confusion matrices for every classifier; from those, AUC with a bootstrap 95% confidence interval, AUPRC, Brier score, Youden's index, accuracy, sensitivity, specificity, F1 score, Cohen's kappa, positive predictive value, and negative predictive value are all derived, at both the Youden-optimal and a fixed 0.5 cutoff for transparency. Because the top few models by point-estimate AUC on a roughly 2,200-participant test set routinely sit within a fraction of a percentage point of each other (well inside plausible sampling noise), a paired bootstrap test (same resampled test-set rows applied to every model within each of 1,000 draws) additionally compares the top-ranked model against every other model and flags which ones are not statistically distinguishable from it at the 95% level; that flag is carried through every subsequent table and figure that ranks models, so the results never present a coin-flip-sized margin as a definitive "best" algorithm. Calibration curves confirm whether predicted risk lines up with observed MetS prevalence. Interpretability comes last: SHAP supplies summary bar plots, beeswarm plots, dependence plots (including a dedicated panel comparing all five metals on a common scale, regardless of each metal's individual rank), force plots, and decision plots to quantify each feature's impact; permutation importance, computed independently across the top several models, cross-checks that ranking for agreement rather than relying on a single model's SHAP output alone.

Four further checks round out the evaluation. A decision-curve analysis compares each model's net clinical benefit against treat-all and treat-none baselines across a range of risk thresholds, since discrimination alone does not establish that a model would change a clinical decision for the better. A subgroup analysis recomputes AUC with bootstrap confidence intervals within sex, race/ethnicity, and age-tertile strata, to check whether performance holds up evenly across the population or concentrates in one group. Four sensitivity analyses on the top-ranked models (adding a diagnosed-diabetes indicator to the predictor set, comparing SMOTENC against class-weighting as the imbalance-handling strategy, comparing the full five-metal panel against a lead-only panel, and comparing the harmonized, medication-adjusted outcome label used throughout this study against the original lab/exam-only definition) show how much the headline results depend on choices that could reasonably have gone the other way. The last of these is the most consequential: because the medication-adjustment clause alone reclassifies 1,201 participants and shifts cohort MetS prevalence by nearly 11 percentage points (section 3.1.2), this sensitivity analysis directly quantifies how much of any reported model's apparent discriminative performance is attributable to that single definitional choice, holding the feature set, models, and cross-validation folds otherwise identical.

3.2. Mendelian Randomization

3.2.1. Study design

This MR study is designed to test whether blood metal exposures, together with any non-metal factor the machine-learning phase identifies as a comparably strong predictor, have a causal effect on Metabolic Syndrome risk and its individual components in individuals of predominantly European

ancestry (Fig. 2), drawing on publicly available summary-level GWAS data; Table 3 summarizes the resulting instrument–exposure–outcome structure exposure by exposure.

The exposure list is not fixed independently of the modeling phase; it is anchored to two criteria applied jointly. First, the exposure has to place among the higher-ranked predictors in the SHAP and permutation-importance analysis described in section 3.1.7: in a preliminary run of this exact pipeline on the 11,144-participant analytic cohort described in section 3.1.1, body mass index and age dominated the ranking by a wide margin, blood metals clustered in the middle to lower portion of the ranking, with lead ranking the most consistently important of the five metals across the independently trained models (present in the permutation-importance top 10 for all four leading models) while cadmium showed the sharpest disagreement between its SHAP and permutation-based rankings; alcohol intake fell in that same middle range alongside the metals, while physical activity emerged as one of the strongest predictors in the entire model, third overall behind only body mass index and age. Second, and just as important, the exposure has to have a genome-wide-significant genetic instrument published in the literature. A predictor with a strong association-model signal but no valid instrument cannot be tested by MR at all, and a weaker signal with a strong instrument is still worth testing, because an associational ranking and a causal test are answering different questions: a factor can be predictive without being causal, or causal without being the single strongest predictor in a model trained on observational data. Table 1 lists the resulting candidate exposures against the instrument evidence located for each.

Figure 2. Mendelian Randomization Study Design

Genetic variants as instrumental variables for blood-metal and metabolic exposures on Metabolic Syndrome

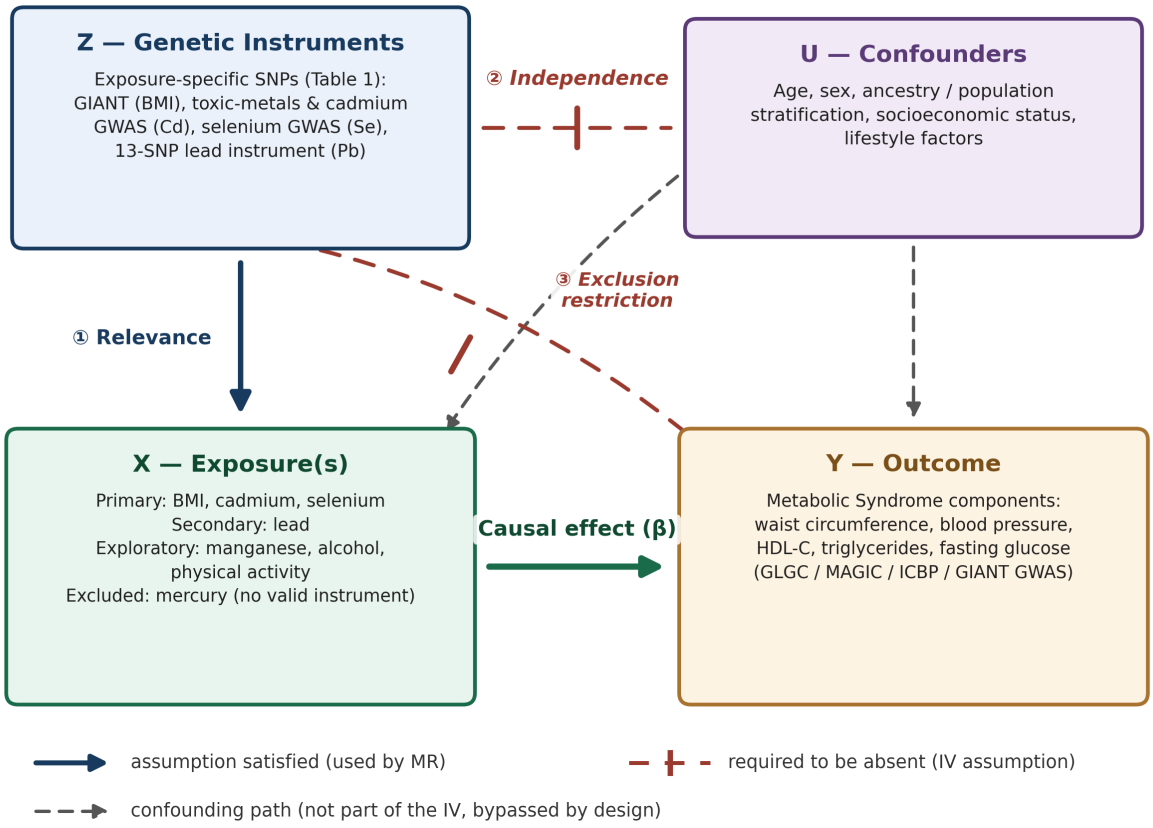


Figure 2. Study design: genetic variants serve as instrumental variables linking each blood-metal or metabolic exposure to the metabolic-syndrome outcome, subject to the three core IV assumptions (relevance, independence, exclusion restriction).

3.2.2. Data sources

GWAS summary statistics are drawn from the IEU OpenGWAS project (https://gwas.mrcieu.ac.uk/) and, where a trait is not indexed there, directly from the original publication's supplementary data. Independent datasets drawn from populations of matched ancestry are selected wherever possible to limit confounding in the genetic associations, and a two-sample MR design is implemented separately for each exposure–outcome pair.

For the metals, exposure instruments come from genome-wide meta-analyses of blood metal concentrations: cadmium and manganese from Ng et al.'s toxic-metals GWAS and the multi-element GWAS by Moksnes et al. [73,74], cadmium specifically also from the never-smoker cadmium GWAS by Borné et al. [75], selenium from Cornelis et al.'s GWAS of blood selenium concentrations [76], and lead from the same genome-wide meta-analyses of blood and erythrocyte metal concentrations, no larger

dedicated blood-lead GWAS being available at the time of writing [73,74]. Mercury is the one metal for which no genome-wide-significant instrument currently exists in the published literature (the largest GWAS of blood mercury to date, in pregnant women and children of European ancestry, identified suggestive but not genome-wide-significant loci [77]), so mercury is not carried into the primary two-sample MR analysis. It will be added only if a qualifying instrument becomes available during the study period, using the same protocol as the other four metals; this decision is revisited explicitly in section 3.2.3. For the non-metal exposure(s) that the modeling phase flags as comparably important, body mass index draws on the GIANT consortium meta-analysis [78,79], and alcohol consumption and physical activity, both retained following the modeling phase, with physical activity emerging as one of the strongest non-metal predictors overall (third of 33 predictors by SHAP magnitude, behind only body mass index and age), draw on the GSCAN consortium [80] and the UK Biobank accelerometer-derived physical-activity GWAS [81] respectively.

On the outcome side, because a dedicated, adequately powered GWAS of the composite MetS phenotype is not consistently available across ancestry-matched cohorts, the primary outcome data are drawn from GWAS of the individual MetS components instead: lipids from the Global Lipids Genetics Consortium, fasting glucose from the MAGIC consortium, blood pressure from the International Consortium of Blood Pressure GWAS or UK Biobank, and waist circumference from GIANT, run as parallel analyses rather than pooled into a single composite outcome. This avoids diluting a heterogeneous phenotype and keeps the results interpretable against the same clinical criteria used to define MetS in the machine-learning phase.

3.2.3. Genetic instrument selection

Instrument SNPs are normally selected at the genome-wide significance threshold ($P < 5 \times 10^{-8}$) for each exposure; if fewer than 15 independent loci clear that bar for a given exposure, a relaxed cutoff of $P < 5 \times 10^{-6}$ is applied instead [82,83]. LD clumping ($r^2 = 0.001$, window = 10,000) using an ancestry-matched reference panel retains only the lowest-p-value SNP at each locus [49,84], and SNPs with potential pleiotropic effects are flagged and removed via the PhenoScanner database [85]. Instrument strength is checked with the F-statistic ($F = \beta^2/SE^2 > 10$) to rule out weak-instrument bias for each exposure individually; this is also the point at which mercury is reassessed: should a future GWAS update supply a genome-wide-significant mercury instrument that clears this bar, mercury is added to the exposure panel using the same protocol as the other four metals, and this addition would be reported as a protocol amendment rather than folded silently into the original analysis. During harmonization (action = 2), palindromic SNPs with intermediate allele frequencies (0.40–0.60) are excluded, along with any SNP whose minor allele frequency falls below 0.01, to keep rare-variant noise out of the estimates [84,86].

3.2.4. Statistical analysis

Causal effects are assessed with several complementary MR methods: inverse variance weighted (IVW), simple mode, MR-Egger regression, weighted median, and weighted mode, applied separately to every exposure–outcome pair described above. IVW serves as the primary estimator, combining SNP-specific Wald ratios under the assumption that no pleiotropy is present [87]; when heterogeneity and pleiotropy

are genuinely absent, maximum likelihood (ML) produces equivalent point estimates with tighter standard errors [87]. MR-Egger, applied under the Instrument Strength Independent of Direct Effect (InSIDE) assumption, catches directional pleiotropy through its intercept term [88]. Weighted median remains valid even if up to half the instruments turn out to be invalid [89,90], and weighted mode adds robustness against certain InSIDE violations that MR-Egger alone might miss.

Heterogeneity in the IVW estimates is assessed via Cochran's Q, and in MR-Egger via Rucker's Q [91]. The I^2 statistic quantifies how much of that heterogeneity matters (below 25% counted as low, 25–75% moderate, above 75% high) [92], and the MR-Egger intercept provides a further check for pleiotropy [88].

Several sensitivity analyses back this up. MR-PRESSO flags and removes outlier SNPs showing pleiotropic effects [93]; RadialMR, Cook's distance, and Studentized residuals help identify any variants exerting outsized influence on the estimates [94]; and a leave-one-out analysis checks whether the results hinge on any single SNP [84].

Because IVW is particularly sensitive to pleiotropy, alternative estimators are also run for comparison: MR-RAPS, MR-Lasso, constrained maximum likelihood, contamination mixture models, and mode-based methods [95-97]. The Steiger directionality test confirms which way the causal arrow points for each exposure–outcome pair, a check that matters in particular for body mass index and metabolic syndrome components, where reverse causation (metabolic dysfunction driving weight gain, rather than the reverse) is biologically plausible and needs to be ruled out rather than assumed away [98]. The Mendelian randomization analyses will be reported in accordance with the STROBE-MR guideline [99].

All of this runs in R version 4.3.1, using packages including TwoSampleMR, MendelianRandomization, MR-PRESSO, RadialMR, MRMix, mr.raps, and GMRP, with additional analyses conducted in Stata version 17 via the mrrobust package. Results are reported as odds ratios and beta coefficients with 95% confidence intervals, and associations with $p < 0.05$ are treated as statistically significant (Fig. 2).

Table 1. Candidate MR exposures identified by the machine-learning phase, screened against published genetic instrument availability.

Exposure	AI-phase signal	Genetic instrument	MR status
Body mass index	Strongest predictor by a wide margin (rank 1 of 33 predictors by both SHAP and permutation importance, all models)	GIANT consortium meta-analysis, ~700,000 participants [89]	Primary exposure
Cadmium	Highest of the five metals by SHAP magnitude (rank 10 of 33), but lowest of the five by permutation importance (rank 18 of 33); the two importance methods disagree sharply for this metal	Genome-wide meta-analyses of blood cadmium [93,94]; toxic-metals GWAS [92]	Primary exposure
Selenium	Mid-ranked and the most consistent of the five metals across both importance methods (SHAP rank 12 of 33,	GWAS of blood selenium concentrations, loci at 5q14 and 21q22.3 [96]	Primary exposure

	permutation rank 11 of 33); SHAP-vs-concentration relationship is a flat-then-rising threshold pattern above ~230 µg/L, not the U-shape described in an earlier draft		
Lead	Second-highest of the five metals by SHAP (rank 11 of 33) and the only metal of the five in the permutation-importance top 10 (rank 9 of 33); the most consistently important of the five metals in this analysis, reversing an earlier draft's "lower-ranked" characterization; still the most MR-precedented	13-SNP instrument used in prior lead–CVD MR work [95]	Secondary / cautious inclusion
Manganese	Lowest-ranked of the five metals by SHAP (rank 21 of 33); near-zero by permutation importance as well, with no clear direction in the SHAP-vs-concentration relationship	Two genome-wide-significant loci identified (4q24, 1q41) [92,94]	Exploratory
Alcohol consumption	Mid-ranked and consistent across methods (average daily drinks, ALQ130: SHAP rank 16 of 33, permutation rank 16 of 33)	GSCAN consortium, ~1.2 million participants [90]	Exploratory
Physical activity	One of the strongest non-metal, non-anthropometric predictors in the model (vigorous recreational activity, PAQ650: SHAP rank 3 of 33, third overall behind only BMI and age; permutation rank 6 of 33); vigorous work and moderate recreational activity rank far lower (SHAP ranks 29 and 18)	UK Biobank accelerometer-derived GWAS, 14 loci [91]	Exploratory (weaker instrument)
Mercury	Fourth of the five metals by SHAP (rank 14 of 33) but at noise level by permutation importance (rank 27 of 33, the lowest of the five metals by that method); no genome-wide-significant instrument exists regardless, so this does not affect the exclusion decision	No genome-wide-significant instrument published to date [97]	Not included — no valid instrument

Table 2. Participant flow for the machine-learning analytic cohort.

Step	N remaining	Excluded at this step
0. NHANES 2007–2018 pooled sample	59,842	—

1. Age ≥ 20 years	34,770	25,072 (age < 20)
2. Complete data on all 5 MetS components (waist, blood pressure, HDL, triglycerides, glucose)	13,337	21,433 (missing ≥1 MetS component)
3. At least 1 of 5 blood metals measured	11,144	2,193 (no metals measured)
4. Final analytic cohort (MetS status computed)	11,144	0

Table 3. MR instrumental-variable design (summary).

Instrumental variable	Exposure(s)	Outcome(s)
Genetic variants (metal- and trait-specific instruments, Table 1)	Blood cadmium, selenium (primary); lead (secondary, cautious inclusion); manganese, alcohol consumption, physical activity (exploratory); body mass index (primary, non-metal)	Metabolic syndrome components: waist circumference, triglycerides, HDL-C, blood pressure, fasting glucose (GLGC / MAGIC / ICBP / GIANT, as applicable per component)

4. Hypotheses and Research Questions

4.1. Machine Learning

4.1.1. Hypotheses

H1: Machine learning models trained on NHANES data will predict MetS presence in adults more accurately than traditional regression-based approaches.

H2: Ensemble-based models such as LightGBM, XGBoost, CatBoost, and Random Forest will outperform simpler models like logistic regression or decision trees.

H3: Blood metals, considered jointly, will emerge as meaningful contributors to MetS risk when assessed alongside conventional demographic, anthropometric, and lifestyle variables, though the five metals are not expected to contribute equally, and at least one is expected to rank below several non-metal covariates.

H4: SHAP and permutation-importance analysis will identify body mass index and age as the dominant predictors of MetS in this cohort, with the blood metals contributing smaller but non-negligible signal overall: lead showing the most consistent ranking of the five metals across independently trained models, and cadmium showing the largest disagreement between SHAP and permutation-based importance.

4.1.2. Research questions

Q1: How reliably can machine learning models trained on NHANES data (2007–2018) predict MetS?

Q2: Among the candidate algorithms (logistic regression, LightGBM, XGBoost, CatBoost, Random Forest, Gradient Boosting, SVM, MLP, and others), which provides the best balance of predictive accuracy, sensitivity, specificity, and calibration?

Q3: How much does adding the five-metal exposure panel improve predictive accuracy over models built solely on demographic, anthropometric, and lifestyle variables?

Q4: What do SHAP- and permutation-importance-based feature rankings reveal about each metal's contribution relative to conventional MetS risk factors, and how consistent is that ranking across independently trained models?

4.2. Mendelian Randomization

4.2.1. Hypotheses

H5: Publicly available GWAS datasets will yield specific SNPs associated with blood cadmium, selenium, and lead levels, and with body mass index, that qualify as valid instrumental variables; instruments for manganese, alcohol consumption, and physical activity will be weaker or more provisional, and no qualifying instrument currently exists for mercury.

H6: The genetic variants linked to the primary exposures (cadmium, selenium, body mass index) will show sufficient instrument strength (F-statistic > 10) to support reliable causal inference; the lead instrument, reused from prior lead–cardiovascular disease MR work and included on a secondary, cautious basis, is also expected to clear this threshold given its established precedent, while instrument strength for the exploratory exposures (manganese, alcohol consumption, physical activity) is expected to be more variable.

H7: Meaningful heterogeneity and potential pleiotropy will be present across the SNP instruments, but robust MR methods (MR-Egger, weighted median, MR-PRESSO) should mitigate the resulting bias.

H8: Elevated cadmium and selenium exposure will be associated with higher risk of MetS overall and of individual components such as elevated blood pressure, impaired fasting glucose, and dyslipidemia; the association for lead is expected to be at least as strong and consistent, in keeping with its being the most consistently ranked of the five metals in the machine-learning phase, though its MR estimate is interpreted more cautiously here because it relies on a genetic instrument reused from prior lead–cardiovascular disease research rather than a metal-specific discovery GWAS.

H9: Exposures ranked highest by the machine-learning phase (body mass index in particular) will show the strongest and most consistent causal estimates in the MR phase, while an exposure with a weak association-model signal but a strong genetic instrument, such as manganese, will help distinguish a genuinely non-causal association from one the modeling phase simply under-detected.

4.2.2. Research questions

Q5: How many genome-wide significant SNPs qualify as valid instruments for each candidate exposure (cadmium, selenium, lead, manganese, body mass index, alcohol consumption, physical activity)?

Q6: Do these SNPs satisfy the three core MR assumptions (relevance, independence, and exclusion restriction) for each exposure?

Q7: How much heterogeneity exists among the SNP-specific causal estimates for each exposure–outcome pair, as assessed via Cochran's Q and I^2?

Q8: Do the MR-Egger intercept and MR-PRESSO tests show evidence of directional pleiotropy influencing any of the exposure–outcome associations?

Q9: Does the MR analysis support a causal association between any of the candidate blood-metal exposures and MetS risk, and if so, does the ranking of causal effect sizes across metals resemble the ranking produced by the machine-learning phase?

Q10: For exposures excluded from the primary analysis due to instrument limitations (mercury) or included only as exploratory (manganese, alcohol consumption, physical activity), does subsequent evidence (either an updated GWAS or the sensitivity analyses in section 3.2.4) change their classification?

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